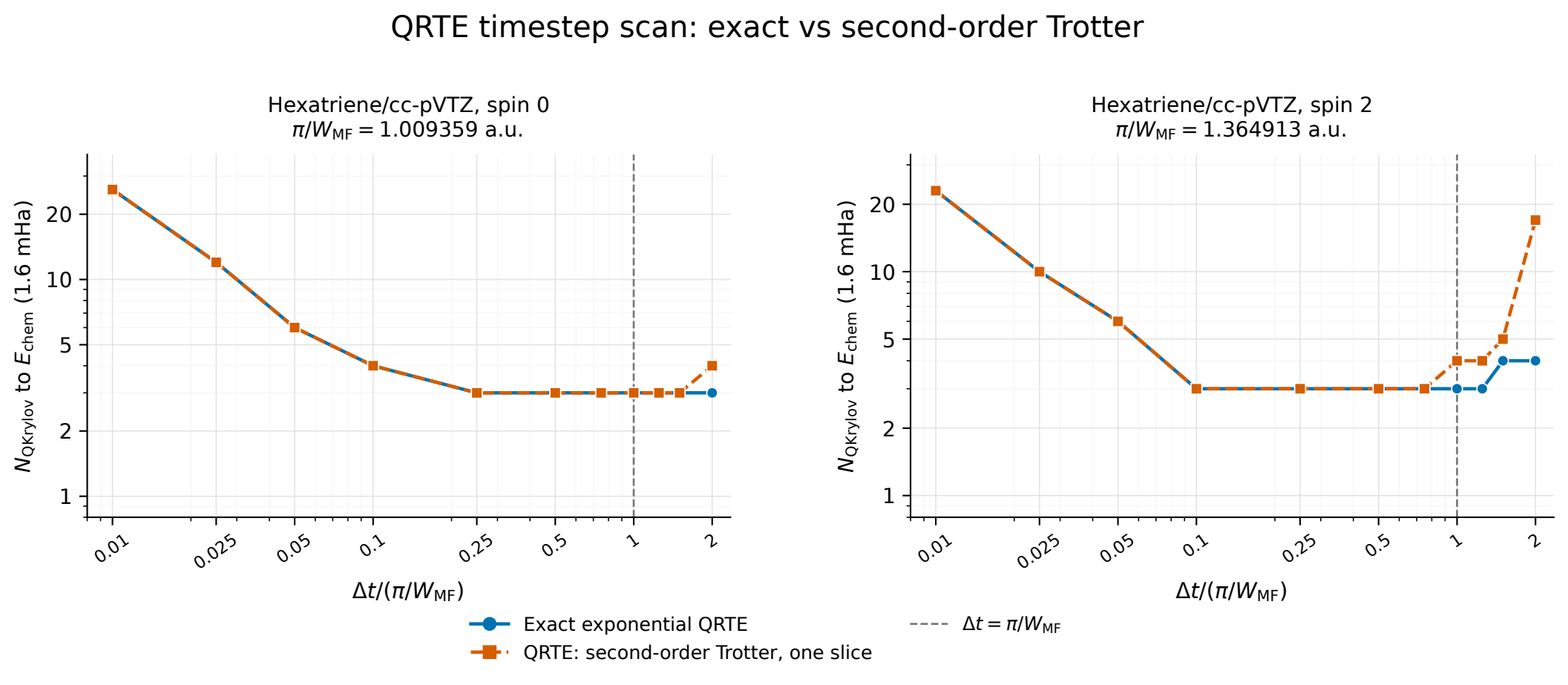
QRTE timestep scan: exact vs second-order Trotter
Hexatriene/cc-pVTZ, spin 0
$\pi/W_{\mathrm{MF}} = 1.009359$ a.u.
Hexatriene/cc-pVTZ, spin 2
$\pi/W_{\mathrm{MF}} = 1.364913$ a.u.
N_{QKrylov} to E_{chem} (1.6 mHa)
20
10
5
2
1
0.01
0.025
0.05
0.1
0.25
0.5
1
2
$\Delta t/(\pi/W_{\mathrm{MF}})$
Exact exponential QRTE
QRTE: second-order Trotter, one slice
$\Delta t = \pi/W_{\mathrm{MF}}$